Problem Set 2

Starter

1 (E)**.** Let G be a group and H, K subgroups of G. Show that $H \cap K$ is a subgroup of G.

2 (E)**.** Show that the set $S(\mathbb{N})$ of invertible functions $\tau : \mathbb{N} \to \mathbb{N}$ is a group under composition. Show that the subset $S^{\text{fin}}(\mathbb{N})$ of invertible functions $\tau : \mathbb{N} \to \mathbb{N}$ such that there is some $n \geq 1$ with $\tau(m) = m$ for all $m > n$ is a subgroup of $S(\mathbb{N})$.

3 (E)**.** Let G be an abelian group. Let $x, y \in G$ be elements whose orders are coprime. Show that the order of xy is $|x||y|$.

4 (L-M)**.** Let G be a group and H, K subgroups of G. Let $HK = \{hk : h \in H, k \in K\}$.

(i) Show that the union $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.

(ii) Give an example where HK is not a subgroup of G. Justify your answer.

(iii) Show that if G is abelian then HK is a subgroup of G.

Advanced

5 (L-M)**.** Consider a dihedral group D_n.

(i) If $n = 2k$ is even and $n \geq 4$, show that $z = r^k$ is an element of order 2 which commutes with all elements of D_n. Show also that z is the only nonidentity element of D_n which commutes with all elements of D_n.

(ii) If n is odd and $n \geq 3$, show that the identity is the only element of D_n which commutes with all elements of D_n.

6 (L-M)**.** Suppose that a finite abelian group does not contain elements of order 2. Show that the product of all elements in G is 1.

7 (L-M)**.** Suppose that a finite abelian group contains an element of order 2. Show that the product of all elements in G equals the product of all elements of order 2.

8 (H)**.** Let G be a finite group, and let $\psi(G) = \sum_{g \in G} |g|$. Prove that:

(a) $\psi(G)$ is odd.

(b) There does not exist a finite group G satisfying $\psi(G) = 2|G|$.

(c) There does not exist a finite group G satisfying $\psi(G) = 3|G|$.

Source: Proposition 2.6, `https://arxiv.org/abs/2210.16666v1`